



**Amirkabir University of Technology
(Tehran Polytechnic)**

Department of Electrical Engineering

Report for the Electronics 1 project
Phase 2

12 V Regulated Power Supply

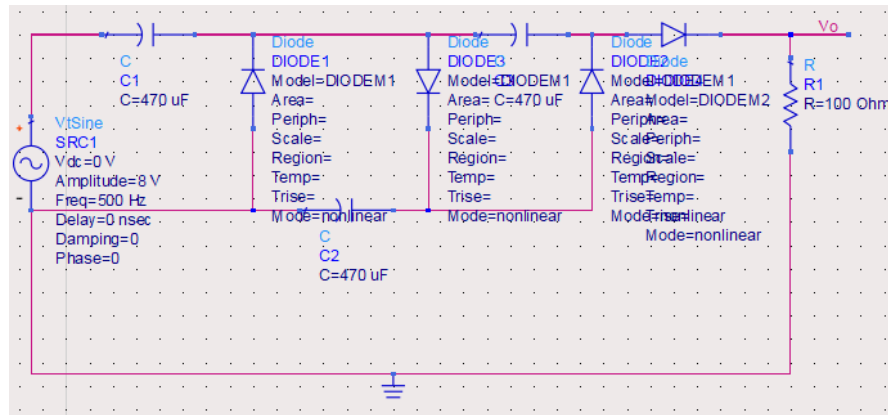
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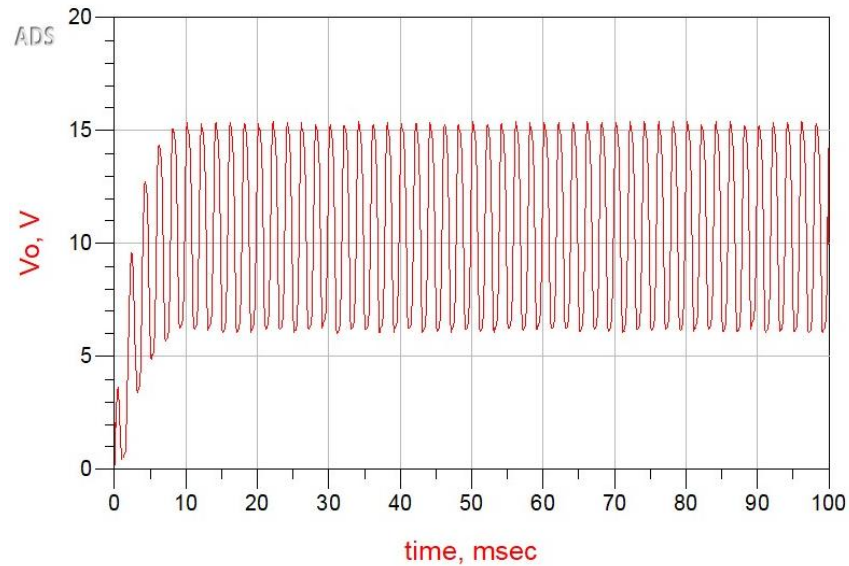
Spring 2023

In the second phase of the project, we are required to design a circuit that, with an input voltage of 8 V amplitude and 500 Hz frequency, produces a constant 12 V output with a maximum ripple of 0.05 V.

In this section, given that the output voltage is higher than the input voltage, we use a voltage multiplier circuit. In this circuit, the output resistance is 100 ohms as stated in the project description. Also, the larger the capacitor value, the better. Therefore, as in the first phase, we use the largest possible capacitance, i.e., 470 μF , and we configure the multiplier circuit as follows:

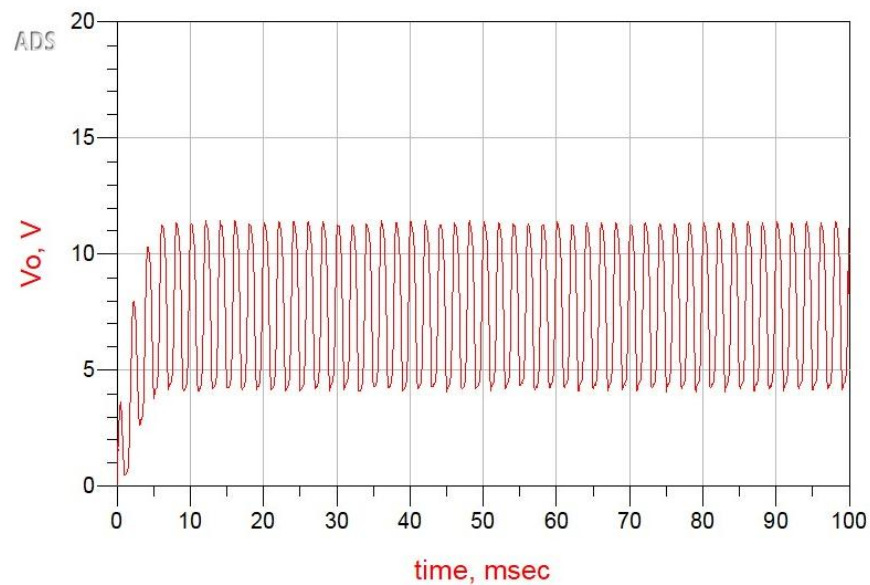


In this case, the output is as follows:

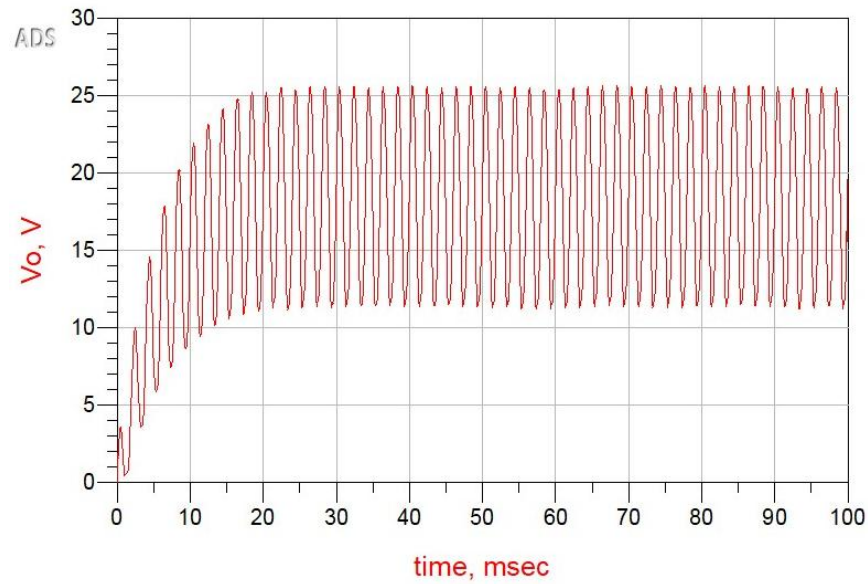


As expected, the output has approximately doubled. But we want a maximum output voltage of 12 V. So we need to make a change in the voltage multiplier circuit. We apply this change by connecting a Zener diode in series with the diodes placed in the circuit, in the opposite direction of the diodes. In this project, we can use a Zener diode with a Zener voltage of 5.1 V and a Zener diode with a Zener voltage of 12 V. Now we try both:

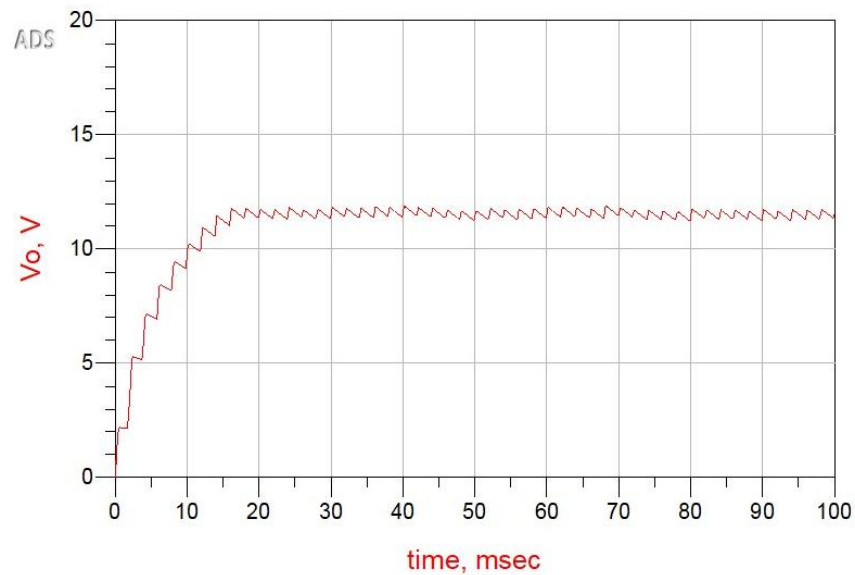
Zener voltage 5.1 V:



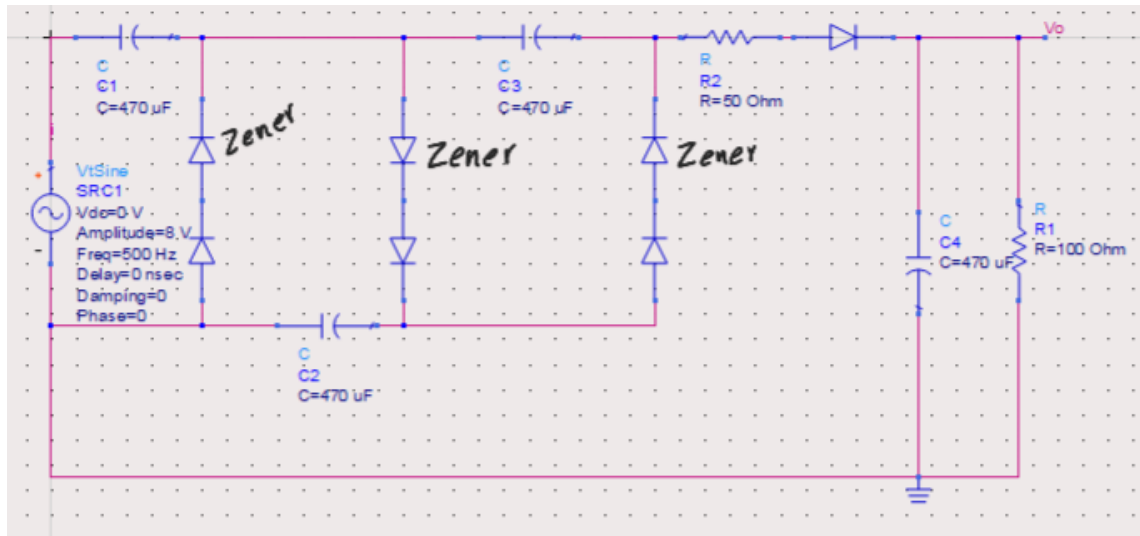
Zener voltage 12 V:



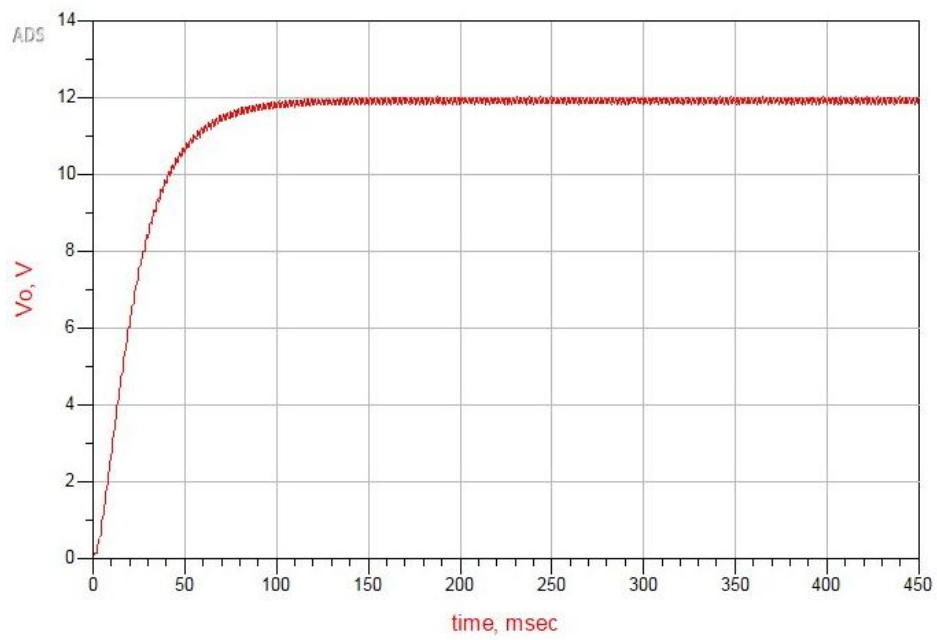
Therefore, the 5.1 V Zener diode is more suitable. Now, to reduce the ripple, it is enough to use a capacitor at the output. The larger the capacitor, the lower the ripple, so considering our limitation, we place a $470\text{ }\mu\text{F}$ capacitor. In this case, the output is as follows:



We still have not reached the desired ripple. To reduce the ripple, we place a resistor in series with the last diode in the circuit and determine its value by testing the output. As a result, we have:



And its output is as follows:



From a closer view:

